Mechanism-aware diagnostics reveal cross-substrate extrapolation failure in CO□ cycloaddition yield prediction

2,490 Reaxys reactions · five epoxides · five catalyst system classes · four-tier evaluation · three evidence lines

1 Data governance

n = 2,490 reactions

5 epoxides (PO, ECH, SO, CHO, IGE)

5 catalyst classes

1970–2024 (Reaxys)

yield ∈ [0, 100]%

81.9% > 70% (right-skewed)

2 Four-tier ladder

(a) 5-fold GroupKFold
groups = catalyst_type

(b) LOMO
groups = mechanism

(c) LOSO
groups = substrate

(d) External holdout

3 Three converging lines

SHAP CI: CHO [−25.61, −24.68] vs PO [+3.60, +3.73] (gap 28.28)

KL: CHO–terminal 14.35 vs LOMO 2.59 (×5.5)

Permutation v2/v3: genuine on terminals, ambiguous on CHO (p=0.78)

4 Deployable action

CHO × organic_base
n = 9, μ = 0.588
single-experiment target

Deployment MAE > R^2
4-seed MAE 0.134 ± 0.007
95% CI [−0.275, +0.537]
cannot reject $R^2 = 0$

$R^2 = +0.228 \pm 0.008$	$R^2 = -0.816 \pm 0.242$	$\Delta R^2 = +1.067 \pm 0.242$	$R^2 = +0.131 \pm 0.128$
LOSO (cached)	LOSO (honest per-fold)	Leakage gap	External holdout

Data: cleaned.csv · bootstrap_substrate_shap_ci.csv · loso_kl_divergence.csv · permuted_feature_ablation_v2/v3.csv · external_holdout/multiseed_v2_summary.csv · n = 2,490 · CC BY 4.0